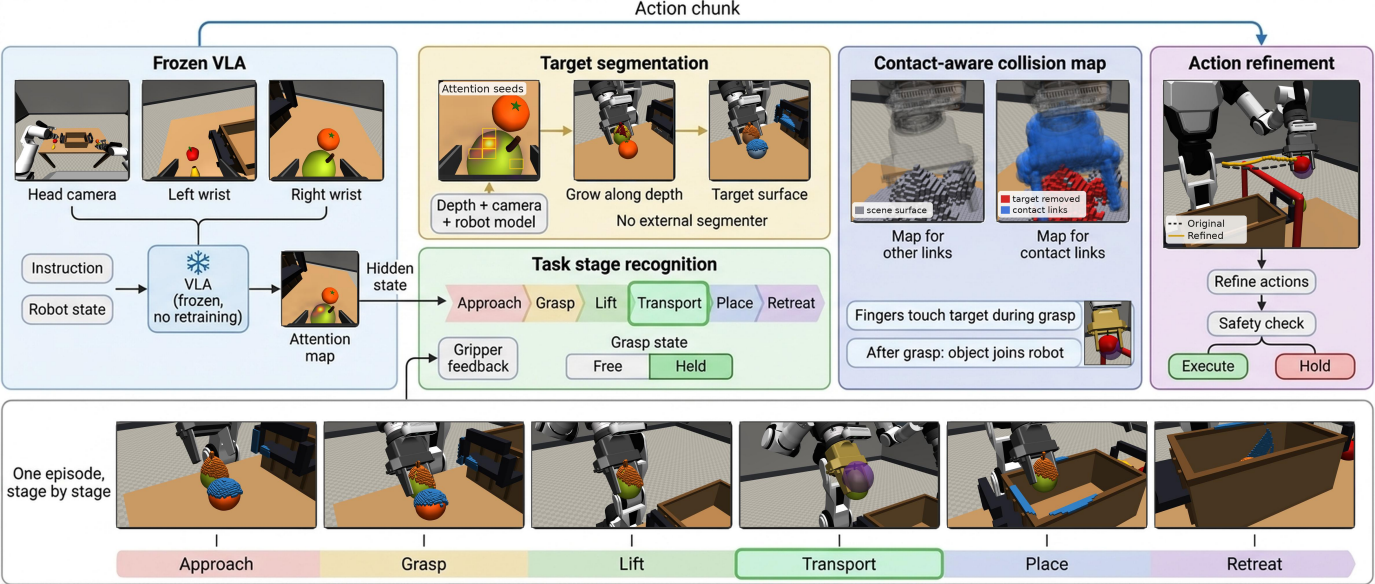

Action chunk
Frozen VLA
Head camera
Left wrist
Right wrist
Instruction
Robot state
VLA
(frozen,
no retraining)
Attention
map
Hidden
state
Target segmentation
Attention seeds
Depth + camera
+ robot model
Grow along depth
Target surface
No external segmenter
Task stage recognition
Approach
Grasp
Lift
Transport
Place
Retreat
Gripper
feedback
Grasp state
Free
Held
Contact-aware collision map
scene surface
target removed
contact links
Map for
other links
Map for
contact links
Fingers touch target during grasp
After grasp: object joins robot
Action refinement
Original
Refined
Refine actions
Safety check
Execute
Hold
One episode,
stage by stage
Approach
Grasp
Lift
Transport
Place
Retreat